Equipment datasheets D-400 and E-400

Process and mechanical datasheets with general-arrangement sketches and the D-400 nozzle schedule

Document no. FD-V1-ME-DS-001

Sheet 1: D-400 flash drum. Sheet 2: E-400 feed preheater. Each sheet gives the process data in the three HMB cases, the mechanical design data and the ASME VIII Div. 1 thickness, and is followed by its GA sketch (drawn to scale from these data in the Mechanical tab).

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Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

Rev	Date	Description	Originator	Checked	Approved
0	28-OCT-26	Issued for teaching	A. Soottitantawat	—	—

DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-ME-DS-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

Sheet 1 · D-400 flash drum

Process data

Case	Design	Maximum rate	Turndown
Feed (two-phase at N1) kg/h	10,208	12,760	6,380
Vapour to PCV-420 kg/h	1,714	2,147	1,063
Liquid to LCV-400 kg/h	8,495	10,613	5,320
Operating pressure kPa g	102.7	102.7	102.7
Flash temperature °C	49.2	49.2	49.2
Vapour / liquid density kg/m³	4.79 / 605.4	4.79 / 605.4	4.79 / 605.4
Vapour fraction at the inlet (mass)	0.170	0.170	0.170

Mechanical design

Item	Value
Code	ASME VIII Div. 1
Size	1600 mm ID × 4800 mm T/T, 2:1 ellipsoidal heads, skirt; 10.72 m³
Design P / T / MDMT	450 kPa g (+ 30.9 kPa liquid head) / 150 °C / 0 °C
Calculated MAWP (hot, corroded)	684 kPa g (stamp 450 kPa g @ 150 °C)
Material	SA-516-70 shell and heads; SA-105 / SA-106-B nozzles; internals SS 316
CA / joint efficiency	3 mm / 0.85 (spot RT, assumed)
Required thickness (UG-27 / UG-32(d)) + CA	shell 6.30 mm, head 6.29 mm
Nominal thickness	shell 8 mm, heads 10 mm (min. 8.5 mm after forming)
Hydrotest UG-99(b)	585 kPa g at the top
Levels (above BT)	LSLL-417 5 % = 0.45 m; LAL 20 % = 0.90 m; NLL 50 % = 1.80 m; LAH 80 % = 2.70 m; LSHH-416 90 % = 3.00 m
Internals	half-open-pipe inlet device on N1; mesh pad 150 mm at 4.50 m; vortex breaker on N3
Weights	empty 3.29 t, operating 5.81 t, test 14.01 t
PSV	PSV-400 set 450 kPa g on the vapour line (FD-V1-PE-RLB-001)

Nozzle schedule

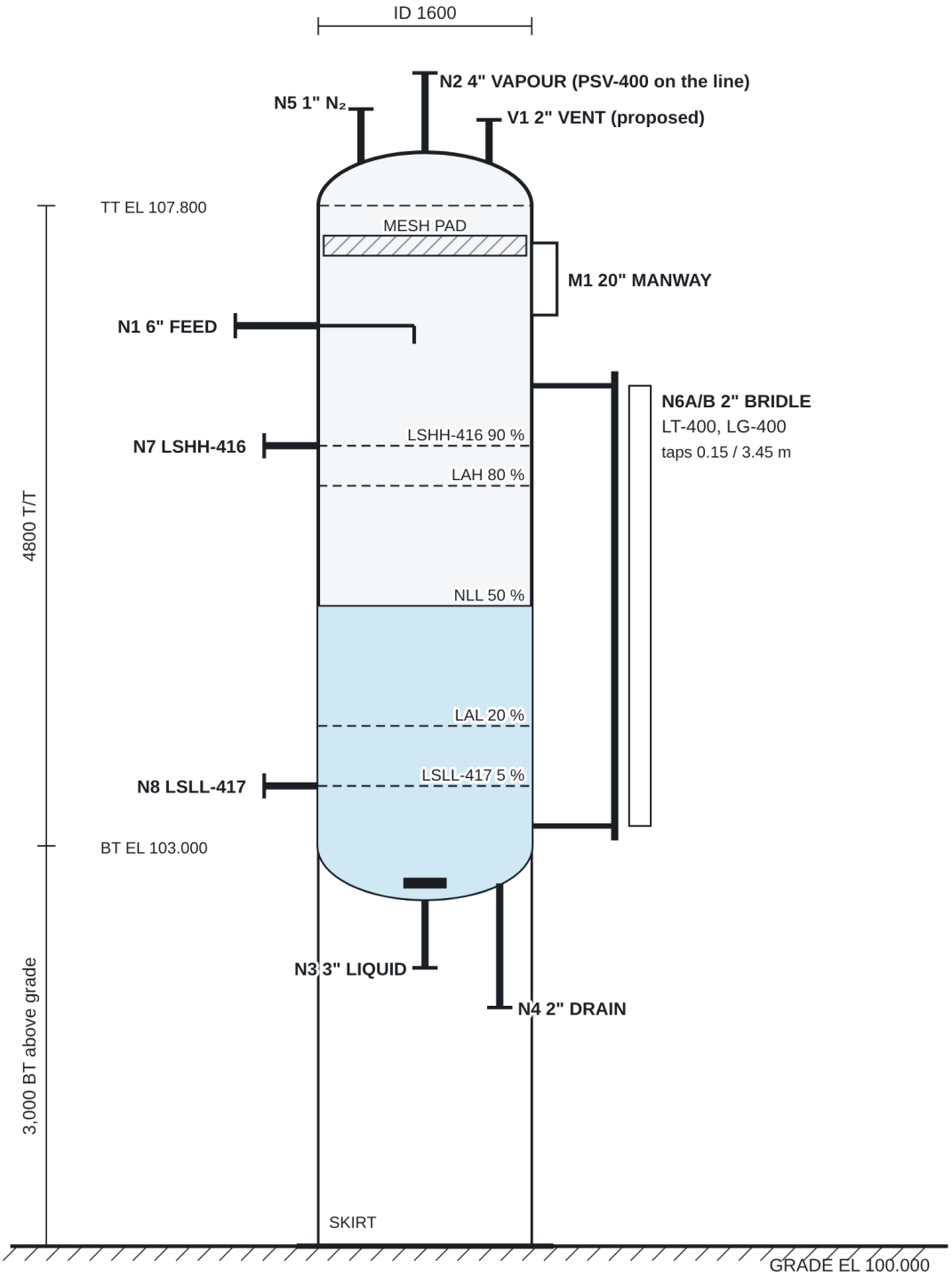
Nozzle	Service	Size in	Elevation m (BT)	Position	Line	On the P&ID	Source
N1	Feed inlet (two-phase) with half-open-pipe inlet device	6	3.90	shell, 270°	6"-P-4004-A1A	FCV-400 → HV-401 → D-400	Design tab nozzle 6" ($\rho v^2 \leq 1500$ Pa)
N2	Vapour outlet (PSV-400 branch on the line)	4	top head	top head, centre	4"-PV-4021-A1A	4"-PV-4021-A1A	Design tab ≥ 3" ($\rho v^2 \leq 3750$ Pa); line 4"
N3	Liquid outlet with vortex breaker	3	bottom head	bottom head, centre	2"-P-4003-A1A	2"-P-4003-A1A	Design tab 3" ($v \leq 1$ m/s)
N4	Drain (HV-415)	2	bottom head	bottom head, off-centre	2"-CD-4051-A1A	HV-415 NC to CD	P&ID
N5	Nitrogen blanket (PCV-430)	1	top head	top head, 315°	1"-N-4031-A1A	1"-N-4031-A1A	P&ID
N6A/B	Level bridle LT-400 / LG-400 (top / bottom connections)	2	3.45 / 0.15	shell, 90°	—	LT-400, LG-400	Design tab (taps outside the span)
N7	LSHH-416 (SIS, own nozzle)	2	3.00	shell, 45°	—	LSHH-416	P&ID, Design tab
N8	LSLL-417 (SIS, own nozzle)	2	0.45	shell, 135°	—	LSLL-417	P&ID, Design tab
M1	Manway (mesh pad passes through, split pad)	20	4.25	shell, 0°	—	—	Design tab 450–600 mm (typical)
V1	Vent / steam-out / purge connection (not on the P&ID)	2	top head	top head, 45°	—	not shown	proposed (query)

D-400 general arrangement (sketch, not for construction)

D-400 FLASH DRUM · GA SKETCH (not to be used for construction)

Design 450 kPa g / 150 °C · MDMT 0 °C · SA-516-70 · CA 3 mm · shell 8 mm, heads 10 mm

Elevations: m above the bottom tangent; levels are % of the LT-400 span



Sheet 2 · E-400 feed preheater

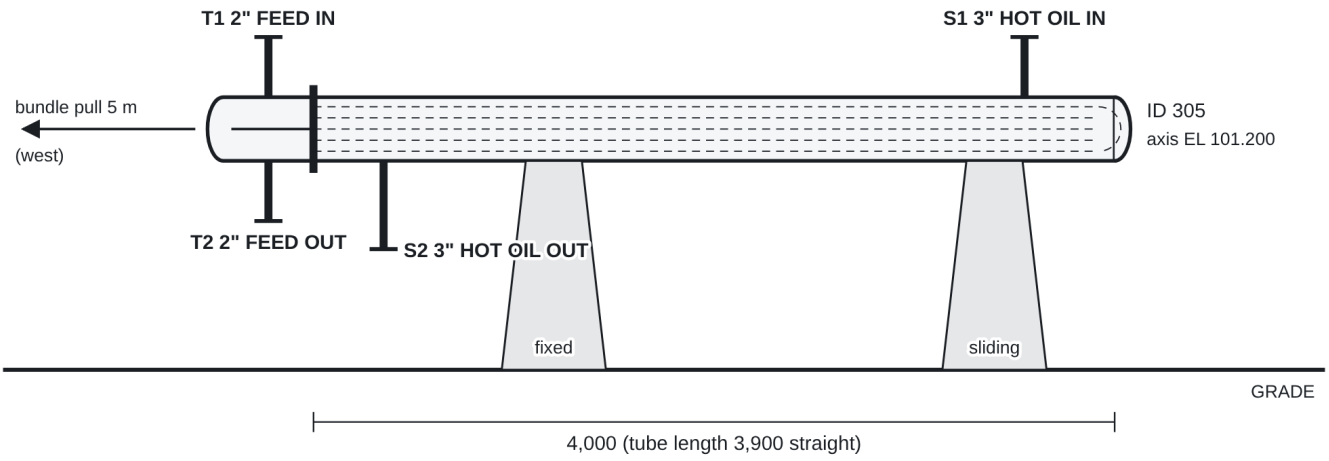
Process data

Case	Design	Maximum rate	Turndown
Duty kW	420.2	525.3	262.6
Shell side hot oil kg/h	10,980	14,692	6,015
Hot oil in / out °C	180.0 / 120.1	180.0 / 124.0	180.0 / 111.7
Tube side feed kg/h	10,208	12,760	6,380
Feed in / out °C	10.0 / 74.0	10.0 / 74.0	10.0 / 74.0
LMTD K / F	108.04 / 0.943	109.97 / 0.949	103.81 / 0.928
Model UA / UA needed kW/K	3.890 / 4.126	4.776 / 5.036	2.530 / 2.725
Area needed m² / excess %	13.75 / 22.2	13.67 / 22.9	13.97 / 20.3

Construction and mechanical design

Item	Value
TEMA type	BEU, U-tube bundle removable to the west
Shell / tubes	305 mm ID × 4000 mm; 72 holes Ø19.05 × 2.11 mm, 3.9 m straight, 2 passes, 25.4 mm 30° triangular
Area	16.80 m² installed; 12.96 m² counter-current at U 300 W/m² K (Design tab, assumed)
Baffles	single segmental, 25 % cut, 300 mm spacing (assumed)
Design P / T	shell 1000 kPa g / 205 °C; tubes 820 kPa g / 205 °C; MDMT 0 °C
Materials	SA-516-70 shell and channel, SA-179 tubes, SA-266-2 tubesheet
Thickness	shell 6 mm, channel 6 mm
Hydrotest	shell 1,300 kPa g, tubes 1,066 kPa g
Fouling	tube 0.00035, shell 0.0002 m² K/W (typical)
Allowable ΔP	tube 40 kPa (model), shell 50 kPa (assumed)
Thermal relief	TSV-400 on the tube side, set 820 kPa g (proposed, query ER-02)
Weights	empty 0.86 t, operating 1.08 t, flooded 1.16 t

E-400 general arrangement (sketch, not for construction)



Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EM-01	D-400 elevation datum	The model's K.Hel = 3 m is the static height of the drum bottom (lowest point) above grade, the plot plan gives the bottom tangent at 3 m. The 2:1 head is 0.4 m deep, so one of them is 0.4 m off.	Static head to LCV-400 changes by ≈ 2.4 kPa; NLL elevation for the fire case changes by 0.4 m.	Fix the datum on the GA: bottom tangent 3 m above grade (plot plan), drum bottom 2.6 m; the model keeps 3 m.	OPEN
EM-02	"MAWP 450 kPa g" on the P&ID	The vessel calculated with the selected plates (shell 8 mm, heads 10 mm) has a MAWP of 684 kPa g (hot and corroded, top). The P&ID calls 450 kPa g the MAWP; it is the design pressure.	The stamped MAWP and the PSV set must be consistent (UG-134).	Stamp the vessel at the design pressure 450 kPa g @ 150 °C (common practice) and state the calculated MAWP on the datasheet; PSV-400 set 450 kPa g.	OPEN
EM-03	D-400 nozzles larger than the P&ID lines	Nozzle N1 (feed) is 6" and N3 (liquid) 3" in the Design tab sizing; the P&ID lines are 2". Nozzle N2 (vapour) 4" matches 4"-PV-4021-A1A.	A nozzle sized for the duty with a line two sizes smaller moves the restriction into the line.	Align with EP-01 / EP-03 (let-down outlet and bottoms line sizes) and show the reducers on the next P&ID revision.	OPEN
EM-04	E-400 thermal design by the vendor	The P&ID gives only the duty (417 kW); the model gives UA 3.89 kW/K clean. The datasheet assumes U 300 W/m² K (Design tab) and a BEU bundle with 16.8 m² (22 % excess at design).	Area, passes and baffles set the tube-side ΔP that the supply curve relies on (40 kPa).	Issue the datasheet for a vendor rating; ask for the clean / fouled U and the ΔP at the maximum rate.	OPEN

No.	Subject	Query	Why it matters	Proposed answer	Status
EM-05	No vent / purge connection on D-400	The P&ID shows a drain (HV-415) but no vent, steam-out or utility connection on D-400.	The drum must be gas-freed for entry and purged with N_2 before start-up.	Add a 2" vent with blind on the top head (V1) and a utility connection; route the vent to the flare.	OPEN